

Lyapunov Vector Fields for UAV Flock Coordination

Dale A. Lawrence¹
Aerospace Engineering Sciences
University of Colorado
Boulder, CO 80309-0429
dale.lawrence@colorado.edu

Abstract

A Lyapunov function approach is used to create UAV guidance vector fields which are globally stable and provide local curl for loitering in circular paths above a target of interest. This provides a means of coordinating a flock of UAVs which is simple to implement on each vehicle, and simple to generate and communicate. This approach is evaluated on an initial simulation which models UAV translation dynamics and acceleration limits, and partitions control actions into separate longitudinal and lateral components. Vector field tracking performance for the small (1 kg) vehicles considered is shown to be compatible with atmospheric sensing applications of UAVs.

Introduction

Unmanned Aerial Vehicles (UAVs) have the potential to revolutionize atmospheric sensing, enabling distributed in-situ sensing in a variety of applications. Tornado formation could be better understood by an array of UAVs sampling pressure, temperature and wind velocity in pre-tornadic storms. Forest fire evolution could be modeled and predicted using in-situ measurements of convective flow in the volume above the fire. Plumes of toxic emissions from industrial accidents could be tracked by a flock

of UAVs, providing timely public safety information.

All these applications have a common need to sense atmospheric data fields, i.e. multiple sensors throughout a volume of interest. Vehicle requirements vary, but generally capability for locating sensors in areas of interest requires directed flight, as opposed to weather balloons. Cost and safety concerns argue for many small vehicles which can be self-launched and tumble or parachute recovered.

Deploying a flock of UAVs raises difficult challenges in command and control. Conventional vehicles benefit from the skill and intelligence of individual pilots, whether on-board or from remote control [1]. This may be appropriate for some military operations, but is extremely expensive. An obvious alternative is to slave vehicles to a lead vehicle, enabling many UAVs to be controlled by a single pilot. This has been pursued, e.g. [2] to reduce fuel consumption in fighter sorties, a la migrating geese. Interesting nonlinear dynamics occur in tight formation flight, and sophisticated control methods are under development, e.g. [3].

Tight formation flying is not necessary, however, in distributed atmospheric sensing applications. Here, it is appropriate to enforce formation more loosely, enabling individual vehicles to sample areas more freely, with individual guidance that may depend on the data each vehicle obtains. This motivates the term *flock*

¹Member AIAA, Associate Professor

rather than *formation*. Stronger formation control may be appropriate when sampled data is sparse or of low quality, hence a method of varying the authority of formation control in real time is of interest. In addition, it would be desirable for human interaction to be raised from piloting toward a higher, more supervisory level, providing intelligent decision making where automation cannot. Finally, in the small UAVs envisioned, the overall command and control method must be simple enough to implement in real time with limited on-board computation and limited communication bandwidth to centralized command and control resources.

The distributed sensing application stands in contrast to military UAVs [4] which have high value and lethality, and must be kept under tight command and control. This results in extremely complex coordination and cooperation problems [6, 7]. For atmospheric sensing applications, we focus on finding the simplest approach that can provide acceptable levels of data sampling so that the region of interest can be characterized with low cost.

In these applications, loss of some vehicles is acceptable, and the danger of a misguided vehicle to conventional aircraft and people on the ground is minimal due to their small size and low flight speeds. Moreover, for landing, failures, or end of battery life, a “tumble mode” or parachute release could be engaged, resulting in high drag and very low speed descents to ground level. For additional operational safety, vehicles can be restricted to operation well above urban structures, and other UAVs can be avoided by potential field barriers (see below). Interaction with other aircraft could be similarly avoided by transponder communication with the UAV flock.

Control Strategy

Control of each vehicle is proposed based on instantaneous guidance vector commands.

At each iteration in the on-board control loop, guidance vectors are obtained from the sampled data, e.g. data concentration gradients, as well as from the formation commands communicated from human supervisors or from centralized data modeling algorithms. These guidance vectors are weighted and summed to produce a desired vehicle velocity vector. This is compared with the current vehicle velocity, and the velocity error is used to form the basis of delta-V (acceleration) commands to the inner loop control system.

Since highly accurate tracking of the desired vehicle velocity is not needed, an optimization trading off tracking with energy loss with computational complexity would be useful for the mapping between velocity error and delta-V commands. Similarly, the weighting between data and formation guidance vectors could take many forms. Appropriate optimization here will depend on flight speeds, length scales of sensed data variation, vehicle dynamics, communication bandwidths, etc.

In this paper, we discuss only the formation component of the guidance vector in detail; data-dependent guidance vectors are not treated. The formation guidance vector is derived from an on-board vector field, where the instantaneous desired velocity is computed by finding the vehicle location in this field using on-board GPS. Although GPS can also provide actual vehicle velocity, update frequencies are too low (about 1 Hz) for inner loop control. Accelerometer data can be used to supply the higher frequency components of inertial velocity, provided body frame measurements are converted to inertial frame coordinates via the relative attitude (pitch, yaw, roll). Vehicle attitude is also needed to implement the inner loop control (see below). Delta-V commands based on desired and actual vehicle velocity are then constrained to longitudinal and lateral vehicle components, enabling (limited) altitude changes

and turns without large variations in airspeed.

Inner loop control to implement the delta-V commands depends strongly on the particular vehicle. In the simplest case, it is envisioned that each UAV will have short period and spiral mode stability from a tail and wing dihedral or sweep. The inner loop control would actuate elevator, throttle, and rudder to provide vertical and horizontal (centripetal) accelerations. Longitudinal airspeed can be controlled by adjusting throttle in response to airspeed deviations from the desired, using propeller speed and throttle setting to derive actual airspeed. Lateral acceleration is provided by actuating rudder and elevator to achieve desired yaw, pitch, and roll angles relative to the desired flight path. Actual pitch and roll angle can be derived from the body frame gravity vector via a 3-axis MEMS accelerometer. Yaw can be measured with a single MEMS rate gyro, correcting gyro drift in yaw by GPS measurements of the flight trajectory. Although more accurate sensing of attitude in small UAVs is becoming feasible with MEMS sensors (see Figure 1), cost and complexity is still an issue. More complex sensing and control schemes, e.g. [11] could be incorporated here if delta-V tracking performance is insufficient. However, extreme maneuvers and complex nonlinear flight control, e.g. [15], are not envisioned.

Guidance Vector Field Design

Formation guidance vectors for a UAV flock in this application should derive from a relatively simple global plan, whose objectives can be rapidly modified and communicated to the flock as centralized processing or human decision making requires. Individual remote piloting of each vehicle is to be avoided, as discussed above, as are complex intelligent or learning control methods [14, 13], and optimal trajectory planning [12, 7]. A suitable approach comes from mobile robotics, e.g. [5], where a poten-

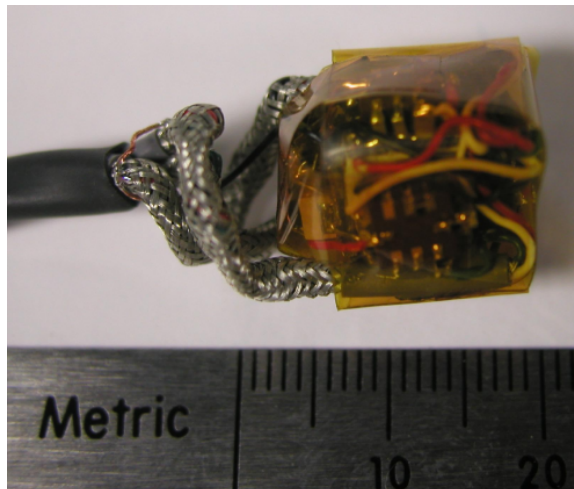


Figure 1: Miniature IMU in development at the University of Colorado contains a suite of 4 MEMS accelerometers, and 3 MEMS rate gyros, all under precision temperature control, in a 14mm cube.

tial function provides a global “plan”, and motion of individual vehicles is easily computed on-board from their position and corresponding local gradient of this potential. This approach is also used for UAV obstacle and collision avoidance [8, 10, 16]. As is, this approach is not convenient for direct air vehicle guidance, unless the vehicle is intended to fly into a target. In the present application, we need the guidance vector fields to have curl (gradients of potential functions do not). With curl, vehicles can stay airborne as they follow the vector field, loitering or circling above areas of interest.

Here, we propose an approach that employs Lyapunov stability theory. but “in reverse”. Instead of constructing Lyapunov functions to analyze stability of a given vector field, we construct families of vector fields to produce desired Lyapunov function behavior: begin with a Lyapunov function with desired formation properties as zero level sets, and with “energy” penalty elsewhere. Then, ensure that the time derivative of this function is non-positive by construc-

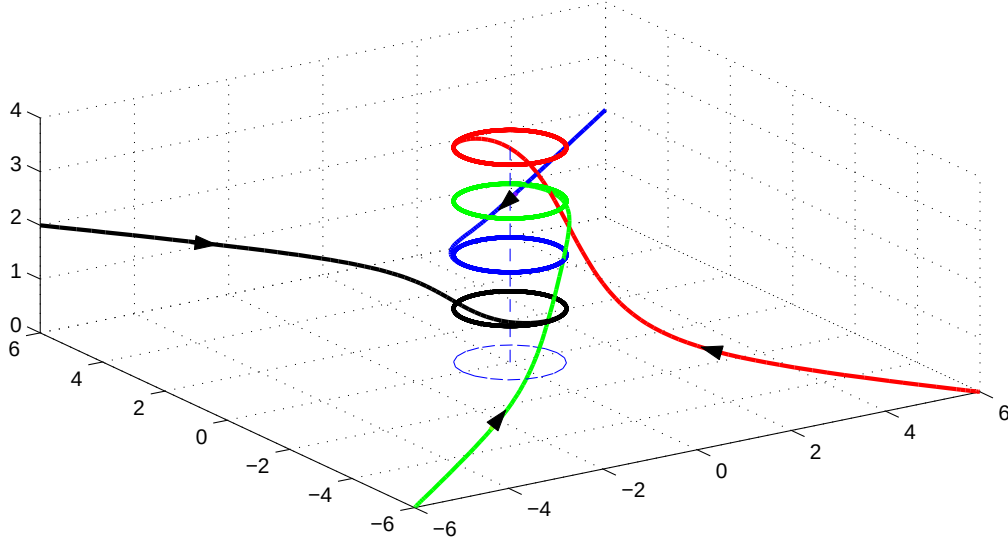


Figure 2: Lyapunov vector field trajectories resulting from four initial positions. All trajectories converge to limit cycles at discrete altitudes (loiter circles) around a region of interest above the target $x = 0, y = 0, z = 0$.

tion of a suitable vector field for the UAV to follow. Figure (2) shows a simple example of a “converge to a target” formation command, where the flock of UAVs are globally attracted to limit cycles orbiting above a location of interest, e.g. a forest fire. When far from the desired location, each UAV is flown directly toward the goal. As the limit cycle is approached, the vector field veers off to smoothly introduce the UAV into a level circular loiter over the source. Each UAV has a unique attractor (limit cycles are at different altitudes) to avoid collisions. In this example, the Lyapunov function V and guidance vector field f are as follows.

$$V(x, y, z) = (x^2 + y^2 - r_d^2)^2$$

$$f(x, y, z) = \begin{bmatrix} -x \frac{r^2 - r_d^2}{r^2 + r_d^2} - y \frac{2rr_a}{r^2 + r_d^2} \\ -y \frac{r^2 - r_d^2}{r^2 + r_d^2} + x \frac{2rr_a}{r^2 + r_d^2} \\ c(h_i - z) \end{bmatrix}$$

This results in the Lyapunov rate of change

$$\frac{d}{dt}V = \frac{-4(r^2 - r_d^2)^2 r^2}{(r^2 + r_d^2)}$$

where $r = \sqrt{x^2 + y^2}$ is the horizontal UAV radius-to-target, and r_d is the desired loitering circle radius. The limit cycles are centered at $(x, y) = (0, 0)$ in this case, with radius $r_d = 1$. The parameter c determines the speed that altitude errors are corrected, and h_i is the altitude of the i^{th} UAVs limit cycle. Note that $\frac{d}{dt}V$ is non-positive, and is zero only on the limit cycle ($r = r_d$) and on the (unstable) equilibrium at $x = y = 0$. Thus [9], the limit cycle is globally attractive, and all UAVs are directed to loiter near a cylinder centered at the desired target position.

Note that simpler attractors (e.g. equilibrium points) are not compatible with sustained flight, hence the vector field must be non-linear, raising difficult analysis and design issues. The

Lyapunov approach provides a relatively simple way to construct guidance laws with rigorous global asymptotic stability properties. Note that the entire flock behavior, beginning at any initial state (UAV position, velocity) is simple to understand and represent by the target location, loiter circle radius, and loiter altitude parameters. These parameters can be easily changed and communicated to the flock in real time as the need arises.

The Lyapunov approach provides a firm foundation for consideration of the inevitable complexities introduced by aerodynamic disturbances, vehicle dynamics, sensor noise, and time varying target parameters. In particular, a wide variation about the “ideal” choice for $f(x, y, z)$ shown above is possible, while preserving the global attraction property. The requirement is that the gradient of V forms an obtuse angle with f , from which a choice for the optimal desired vehicle velocity can be chosen based on current velocity, acceleration limits, etc. Alternatively, significant disturbances or inaccuracy in tracking the desired velocity f can be tolerated.

Such a guidance law is simple to implement on each UAV, and requires very little communication. Estimates for vehicle location are provided periodically by onboard GPS, and combined on-board with Lyapunov vector fields f to determine the desired velocity $\bar{v}_L$, resulting in delta-V commands to the lower level servo loops.

Initial Evaluation

A detailed investigation into the behavior of a UAV flock under Lyapunov vector field guidance has been initiated by simulating the displacement dynamics of a UAV, acted on by forces related to the error between the vehicle velocity and the reference velocity provided by the Lyapunov vector field above. As discussed earlier, forces would be provided by attitude

changes on the vehicle (via control surface deflections) and by changes in the force provided by the propulsion system. For this investigation attitude dynamics were not modeled, although limits on the magnitude and direction of lateral and longitudinal forces were imposed, consistent with small aircraft flight.

Each vehicle was modeled as a point mass $m = 1$ [kg], acted on by the forces described below. In these descriptions, $\bar{v}$ is the vehicle velocity vector, $\bar{w}$ is the prevailing wind velocity vector, $\bar{v}_L = f(x, y, z)$ is the Lyapunov reference velocity vector (normalized to a constant desired airspeed s), and “hats” indicate corresponding normalized (unit) vectors. Desired airspeed s is constant at 10 [m/sec], and vehicle velocity is presumed to remain approximately aligned with the aircraft longitudinal axis.

Longitudinal control force

$$f_{LOC} = k_1 [\hat{v}\hat{v}^T(\bar{w} - \bar{v}) + s\hat{v}]$$

Note that this control force is provided directly by the propulsion system, and is proportional (by the gain k_1) to the difference between the longitudinal component of relative wind and the desired air speed. This control attempts to keep longitudinal airspeed constant.

Lateral control force

$$f_{LAC} = k_2 [(I - \hat{v}\hat{v}^T)(\bar{v}_L - \bar{v})]$$

This control force is proportional (by the gain k_2) to the lateral component of the mismatch between desired (Lyapunov) velocity $\bar{v}_L$ and vehicle velocity $\bar{v}$. This provides the force needed for turning and altitude changes to follow the Lyapunov vector field. Turn rate is limited by saturating this control force.

Drag forces

Separate longitudinal and lateral drag

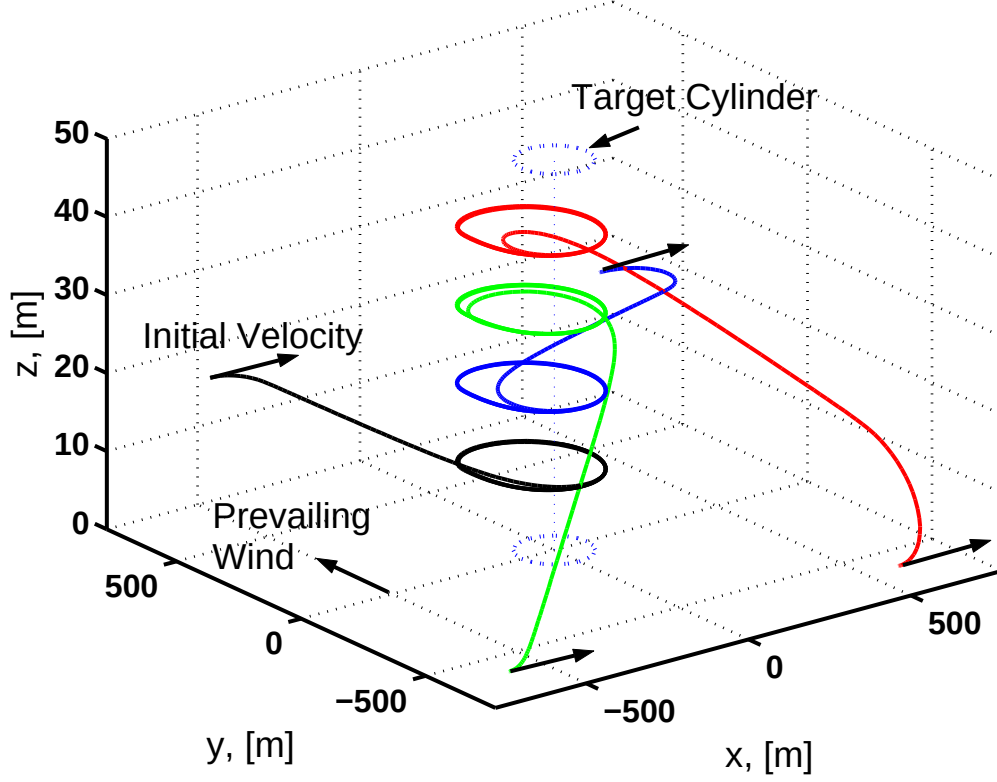


Figure 3: UAV trajectories resulting from lateral and longitudinal control forces attempting to cause tracking of the Lyapunov vector field. Note that the loiter circles are somewhat enlarged, and that the prevailing wind causes the circles to be stretched downwind.

forces are computed based on the corresponding components of the relative wind, using the conventional squared velocity law, with drag coefficients of 0.1 and 1.0, respectively.

Gravitational forces

Airspeed is approximately constant, so lift forces are assumed to balance gravity forces except for the component $f_g = m\bar{g}^T \hat{v}$ in the longitudinal direction. Together with limits on longitudinal control force, this provides a limit on climb rate.

Figure 3 shows a simulation of these forces on four UAVs. Each UAV begins at a different lo-

cation, but all have the same initial velocity of 10 [m/sec] along the positive x direction when the control law is enabled, and the prevailing wind is along the y direction, with a magnitude of 2 [m/sec]. Control gains of 1 [N/(m/sec)] were used for both k_1 and k_2 , with limits corresponding to 0.4 [g] in both lateral and longitudinal directions. The target cylinder has a radius of 100 [m] centered at $x = y = 0$, with separate altitudes corresponding to each UAV of 10, 20, 30, and 40 [m].

Observe that the initial UAV trajectories rapidly bend toward the target circle, climbing or descending at nearly constant rates, as necessary, to approach their respective loiter limit cy-

cles. Tracking of the limit cycles is only approximate, due to the somewhat low control gains on velocity errors near the circles, providing less radial acceleration than needed to stay precisely on the target circle. Also, the prevailing wind distorts the flight path downwind, as one would expect. Better tracking of the Lyapunov trajectory can be achieved with higher control gains (and possibly higher acceleration limits), but as precision tracking is not needed in the applications of interest, this performance seems satisfactory.

The next step in evaluating this approach is to include attitude dynamics on each vehicle, along with control allocations for the control surfaces which attempt to achieve the lateral accelerations demanded by the above guidance law. At this level, optimization of the guidance law relative to vehicle control limitations can be investigated.

Conclusion

Atmospheric sensing applications of UAVs point toward flocks of small, loosely formed vehicles, coordinated by a global plan which can be easily designed and changed as conditions warrant. This coordination plan should be simple and easy to implement on small flight processors. A Lyapunov stability approach was shown to provide a promising means for designing a reference vector field for each UAV to follow. This field enables simple guidance laws to be implemented on each vehicle, with parameters for target location, loiter circle diameter, loiter altitude, etc. that can be easily changed to redirect the entire flock.

Initial simulations incorporating translation dynamics and separate lateral and longitudinal control laws indicate that tracking performance is compatible with small vehicle flight capabilities and the needs of atmospheric sensing applications.

Current research is investigating variations

on the simple limit cycle attractors presented above, with a goal of developing attractors and corresponding guidance laws which are suited to particular distributed sensing application scenarios and particular UAV vehicle capabilities.

References

- [1] A. J. Thurling and K. A. Greene, "An Improved Predictive Algorithm for Time Delay Compensation in UAVs", *Proc. AIAA Atmospheric Flight Mechanics Conf. and Exhibit*, AIAA paper No. 2001-4424, Montreal, Aug., 2001.
- [2] C. E. Hanson, M. J. Allen, J. Ryan and S. R. Jacobson, "Flight Test Results for an Autonomous Formation Flight Control System", *proc. 1st AIAA Tech. Conference & Workshop on UAV, Systems, Technologies, & Operations*, AIAA paper No. 2002-3431, Portsmouth, VA, 2002.
- [3] R. Olfati-Saber and R. M. Murray, "Distributed Structural Stabilization and Tracking for Formations of Multiple Dynamic Agents", *Proc. 41st Conference on Decision and Control*, Las Vegas, NV, Dec., 2002.
- [4] J. L. Hinchman, "Distributed Control: A Path Riddled with Challenges", *proc. 1st AIAA Tech. Conference & Workshop on UAV, Systems, Technologies, & Operations*, AIAA paper No. 2002-4500, Portsmouth, VA, 2002.
- [5] E. Rimon and D. E. Kodischek, "Exact Robot Navigation Using Artificial Potential Functions", *IEEE Transactions on Robotics and Automation*, Vol. 8, No.5, pp. 501-518, 1992.
- [6] I. H. Yang and Y. J. Zhao, "Real-Time Trajectory Planning for Autonomous Aerospace Vehicles Amidst Static Obstacles", *proc. 1st AIAA Tech. Conference & Workshop on UAV, Systems, Technologies, & Operations*, AIAA paper No. 2002-3421, Portsmouth, VA, 2002.
- [7] W. M. McEneaney and B. Fitzpatrick, "Control for UAV Operations Under Imperfect Information", *proc. 1st AIAA Tech. Conference & Workshop on UAV, Systems, Technologies, & Operations*, AIAA paper No. 2002-3418, Portsmouth, VA, 2002.
- [8] H. Van Dyke Parunak, M. Purcell, and R. O'Connell, "Digital Pheromones for Autonomous Coordination of Swarming UAV's", *Proc. 1st AIAA*

Tech. Conference & Workshop on UAV, Systems, Technologies, & Operations, AIAA paper no. 2002-3446, Portsmouth, VA, 2002.

[9] J.-J. E. Slotine and W. Li, *Applied Nonlinear Control*, Prentice-Hall, 1991.

[10] M. M. Polycarpou, Y. Yang, and K. M. Passino, "A Cooperative Search Framework for Distributed Agents", *Proc. IEEE International Symposium on Intelligent Control*, Mexico City, Sept., 2001.

[11] F.-B. Hsiao, K.-W. Lin, W.-T. Lee, W.-Y. Chang, C.-T. Chao, and C.-F. Lin, "The Design of Flight Stability Control System and Constructing Simple Navigation System for a Long Endurance Autonomous UAV", *proc. 1st AIAA Tech. Conference & Workshop on UAV, Systems, Technologies, & Operations*, AIAA paper No. 2002-3469, Portsmouth, VA, 2002.

[12] A. Richards, J. Bellingham, M. Tillerson, and J. How, "Coordination and Control of Multiple UAVs", *Proc. AIAA Conf. Guidance, Navigation, and Control*, Monterey, CA, Aug., 2002.

[13] K. Krishnamurthy and D. T. Ward, "An Intelligent Flight Director for Autonomous Aircraft", *Proc. 38th Aerospace Sciences Meeting and Exhibit*, Reno, NV, Jan., 2000.

[14] Y. Yang, A. A. Minai, and M. M. Polycarpou, "Decentralized Cooperative Search in UAVs Using Opportunistic Learning", *Proc. AIAA Conf. Guidance, Navigation, and Control*, Monterey, CA, Aug., 2002.

[15] Y.-Y. Lin and S.-S. Liu, "Integrated Projection Guidance and Nonlinear Control of Unmanned Air Vehicle", *Proc. AIAA Modeling and Simulation Technologies Conference and Exhibit*, AIAA paper No. 1998-4536, Boston, MA, Aug., 1998.

[16] Y. J. Zhao and D. Rock, "Conflict Alerting Functionality for Autonomous Vehicles", *proc. 1st AIAA Tech. Conference & Workshop on UAV, Systems, Technologies, & Operations*, AIAA paper No. 2002-3420, Portsmouth, VA, 2002.

[17] D. Hristu and K. Morgansen, "Limited Communication Control", *Systems and Control Letters*, Vol. 37, pp. 193-205, 1999.